Milestone 2: Dataset Preparation

Socratic AI Tutor

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What we prepared

Dataset	Used for	Size after preparation
ASSISTments 2009	Learner model (DKT)	282,619 answers, 3,852 students
MathDial	Socratic model fine-tuning	18,541 tutor replies
NCERT Maths + Science	Course pack (study text)	3,261 text chunks, 10 books

All three are public. Raw files are not committed; scripts in data/ rebuild everything.

ASSISTments: learner model data

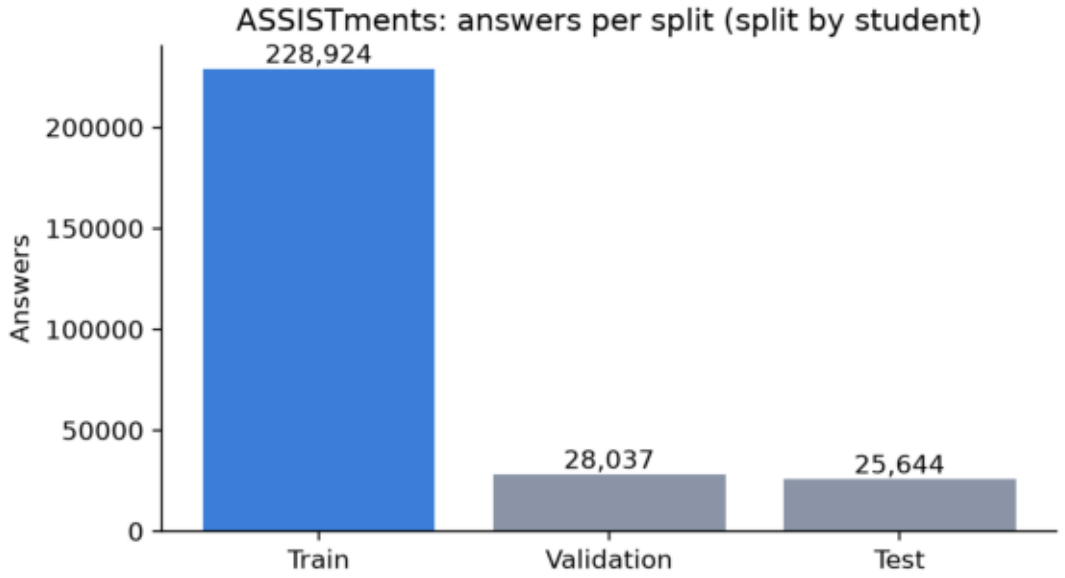
- Raw: 346,860 answers from 4,217 students

- Removed 63,755 answers with no skill label

- Dropped students with fewer than 3 answers (pyKT recipe)

- 149 skills, histories cut into windows of up to 200 answers

- Split by student: 80% train, 10% validation, 10% test



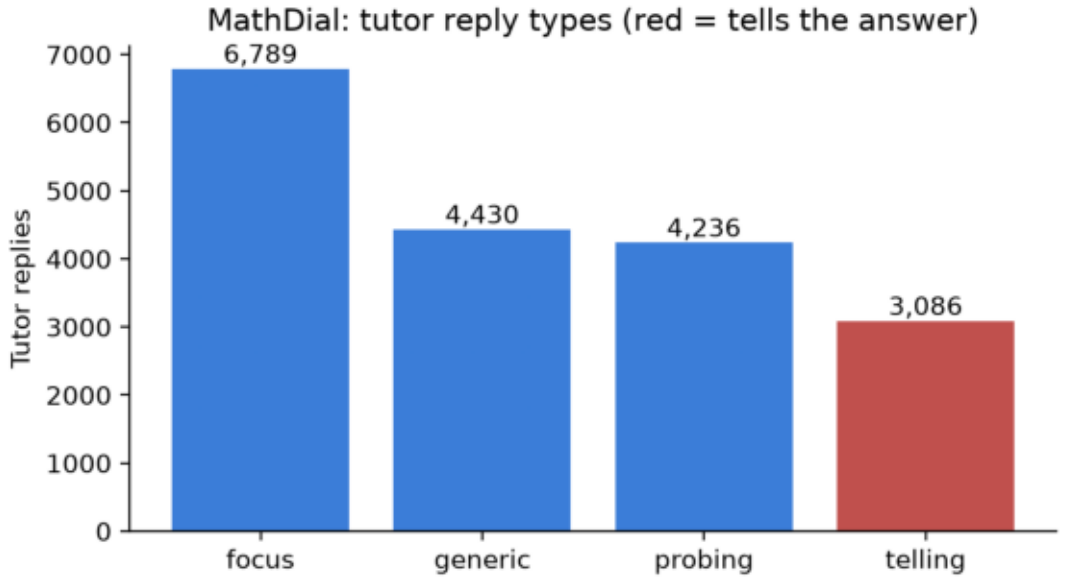
MathDial: Socratic model data

- One example per tutor reply: problem + conversation so far -> next reply

- Each reply keeps MathDial's label: probing, focus, generic, telling

- 'Telling' replies and replies containing the final answer are flagged and not used as training targets

- Validation split by problem, so a problem never sits in both train and validation



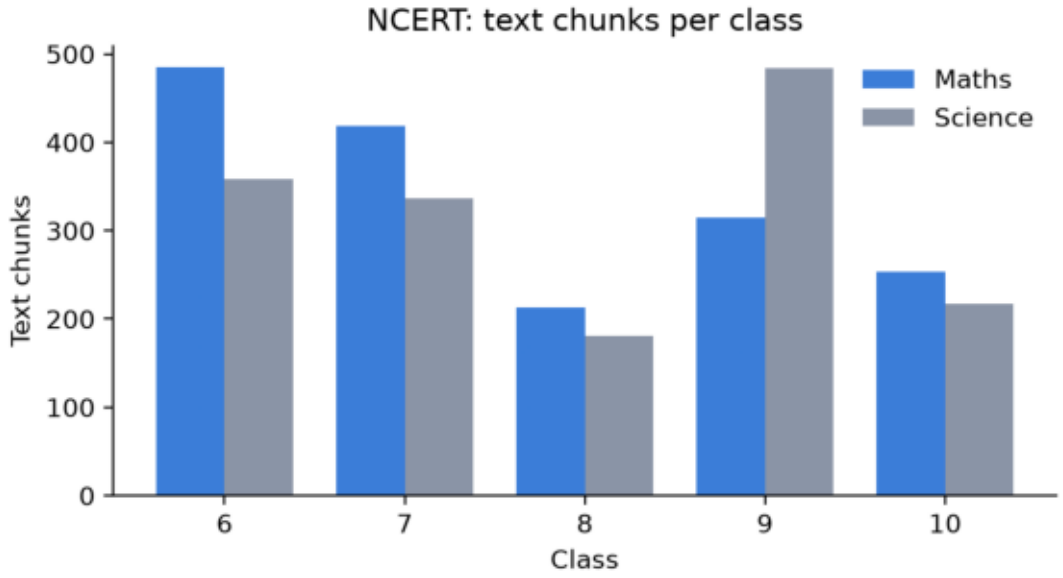
NCERT: course pack text

- Maths and Science, classes 6 to 10 (classes act as levels)

- 2,401 pages cleaned and cut into chunks of up to 250 words

- Each chunk keeps subject, class, chapter and page, so the tutor can point to the exact page

- Text stays local: the books are public but not ours to redistribute



Checks and limits

- Automatic checks pass: no student or problem shared across splits, labels valid, no leaked answers in kept examples

- ASSISTments is US data and has no session information

- 283 of 394 MathDial test problems also appear in train (MathDial's own split)

- NCERT chapter titles are guesses and need a manual pass

Next steps

- Write NCERT concepts, prerequisite links and practice questions

- Train DKT on ASSISTments and compare with the published baseline (AUC 0.754)

- Fine-tune the Socratic model on MathDial plus our NCERT examples

- Label our own test scenarios: backtracking, misconceptions, guardrails